

Qatar-1b

R-1189

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_201011 · created 2026-07-31T05:31:49+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459133.7933 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.026
Transit depth (Rp/Rs)^2	3.24 +/- 0.95 [%]
Orbital Inclination (inc)	81.66 +/- 0.98
Airmass coefficient 1 (a1)	1.018 +/- 0.013
Airmass coefficient 2 (a2)	-0.0095 +/- 0.0058
Scatter in the residuals of the lightcurve fit is	1.27 %
Best Comparison Star	None
Optimal Aperture	3.42
Optimal Annulus	16.25
Transit Duration (day)	0.054 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.026 vs 0.1463 (deviation 1.3σ)
Duration fit vs published	0.054 d vs 0.0692 d (deviation 1.38σ)
Mid-time O–C	0.4 min
Depth SNR	6.9
Residual scatter	1.27 %

Light curve

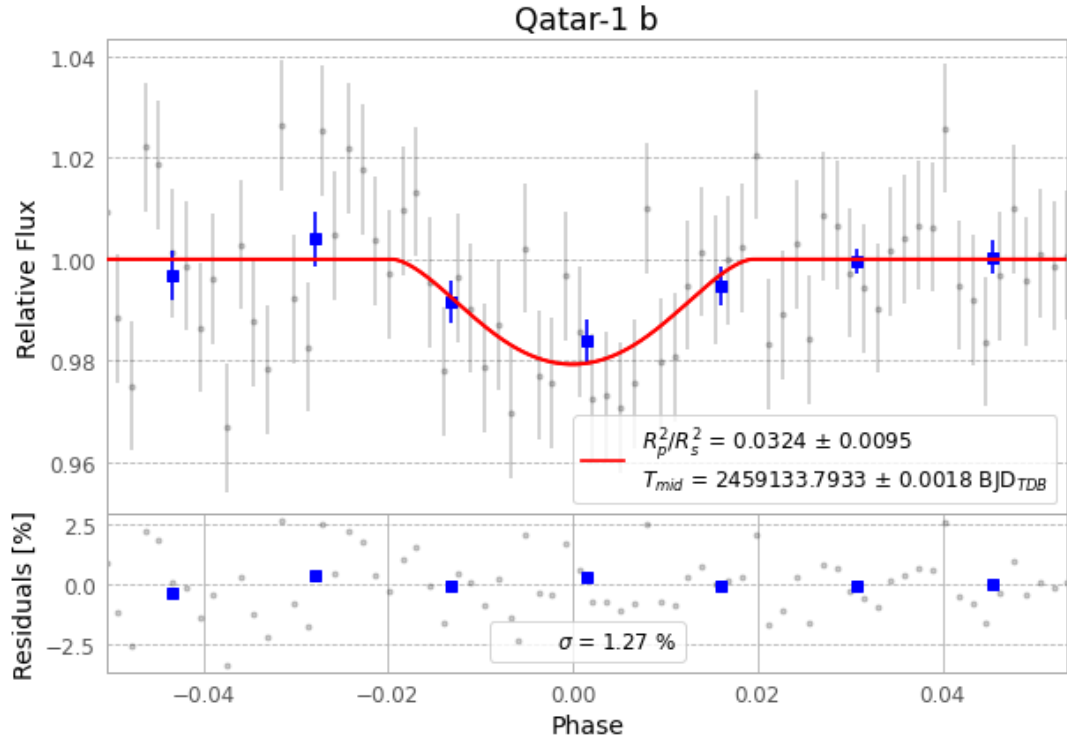


Figure 1 — FinalLightCurve_Qatar-1 b_2020-10-10.png (SHA-256 24c95ecb481a2b7e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 322], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5b55ed4fe3e36eef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), Qatar-1201011051543.FITS → Qatar-1201011084812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-201011.log (b7f8e8e7b402...), FinalLightCurve_Qatar-1 b_2020-10-10.png (24c95ecb481a...), FinalLightCurve_Qatar-1 b_2020-10-10.csv (4fe0ccdd0151...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 05:31Z.